

blueblack

Libraries

These libraries only work with .

Nodes ext.nodes This library extends `'s` functionalities around nodes. NodesOnLine-Q, NodesOnCurve-QNodesOnL
 2 Pic as a node //pic=boolean (default true, initially false) tikz This key allows one to use a pic where usually only
 Node PicturesA Node Picture is a mix of a pic, a matrix and nesting pictures as well as using the fit library while

In this table, means that the functionality is supported by the feature, – means it is not and † means the functionality
 whether the feature can be nested

matrices cannot be nested. Nodes that are |fit|ted don't actually contain the nodes they are fitted to.
 whether one can reference coordinates/nodes outside of a feature

It is not possible to reference a node in the outer picture from inside a matrix. Referencing node from the outer sc
 pictures need |remember picture| to be able to reference a node in the outer picture.

whether one can reference coordinates/nodes from outside of a feature

pictures need |remember picture| to be able to reference a node in the inner picture.

whether the feature can be transformed

A matrix can only be shifted, a |fit|ted node can be transformed but its content will remain where it has been plac

whether settings from outside the feature gets inherited

This is usually something one has tp fight against and can be quite messy to deal with. Only the matrix allows som
 whether a feature can be placed well

A pic doesn't provide any anchors and can't be placed other than at the origin but it can be placed along a path.

Overview The Node Picture shall offer a compromise between a pic and a matrix without having to nest pictures.

Here are some rules and guidelines for using this:

The Node Picture will be typeset as the text of the node.

The baseline of this text is the vertical center of the Node Picture's bounding box. This can be changed with the |ext/n

Only the |text| node part is available. The command may not be used.

A Node Picture is *not* a matrix. It only contains one cell.

Just like with a pic or a matrix, certain settings and values will be inherited by the Node Picture. (It's a gift and a curs

The bounding box of the picture will be evaluated automatically and the result will be used to determine the size of the

You can use a Node Pictures inside a Node Picture. [preamble=ext.nodes, width=3cm] [nodes=draw, do/.code= [roun

Most of the time, all that is needed is to add |/tikz/ext/node picture| to a node and it will be a Node Picture node

Styles //every node picture=Node Picture level This style will be applied for every node that is enabled to be a No

Its parameter is the current Node Picture level. //every outer node picture=Node Picture level This style will be

Its parameter is the current Node Picture level. //every node picture scope=Node Picture level This style will be

Its parameter is the current (outer) Node Picture level.

Settings //node picture=true or false or any of the options below (default true) This is only a key to set keys in th

It is not possible to set the Node Picture's name to an empty value. //name prefix .. Similar to a pic, name prefix

(default 0,0) This applies a transformation so that the coordinate inside the Node Picture lies at the same coordinate in

This also may make it possible to sucessfully reference nodes outside of a Node Picture unless you add further tran

More and Examples A TeX-macro is available for accessing the current Node Picture level. This command returns

Further more, anchors can be added to a Node Picture's node. //add anchor=nc This will add the anchor n to the

A matrix alone is not able to be transformed but as part of this construct, it is possible. As you can see in the nex

Unfortunately, due to how a matrix sets up the naming of its nodes, a name prefix would be applied twice (see [?]).

The example below also shows that all nodes include their own inner sep: The outer node (solid line), the matrix n

The following example is adapted from [?]. [preamble=ext.nodes, arrows.meta, positioning, graphs] [ext/node pict

Nodes on paths When nodes are placed along paths they don't interrupt the path at that place. The decoration ma

This library provides alternatives. These are separated into straight paths, i. e. –, and everything else (including an

Nodes on Lines

//node on line=anchor specification (default { }) tikz This installs a to path that places *one* node along a straight

This allows a node to be placed *on* a straight line without having to use |fill = white| or similar tricks to make the

The optional anchor specification allows to specify the anchors to which the line should connect. It allows one or tw

//nodes on line tikz This is similar to the previous key but allows multiple nodes to be placed on a straight line *if*

It allows *no* anchor specification.

[preamble=ext.nodes, quotes] [inner sep=.15em, circle, nodes=draw, sloped] [ultra thick, ->, ext/node on line] (0,0)

Nodes on Curves The following keys need the intersections and the spath3 [?] library to be loaded. They will not b

Any [/pgf/]outer sep will be ignored.

If you can, use fill=bg color instead of these keys, it will be much faster and easier.

//nodes on curve=to path (default line to) tikz Similar to |nodes on line|, this key allows to have nodes on arbitrar

This is not suitable for paths connecting nodes.

//nodes on curve'=to path (default line to) tikz As above but suitable for connecting nodes.

[preamble=ext.nodes, intersections, quotes, spath3] [ultra thick] (A) at (0, 0) A ; (B) at (3, 0) B ; [red, ->, ext/nod

Automatic placement of nodes The auto key allows automatic placement of nodes along a path segment. This libra

More than left and right Besides |left| and |right| that are provided by the following placement mechanism are pro

|ext/left| will place a node to the left of the direction of the line,

|ext/right| will place a node to the right of the direction of the line,

|ext/above| will place a node towards the direction of the line,

|ext/below| will place a node against the direction of the line,

|ext/west| will place a node towards the left side of the paper,

|ext/east| will place a node towards the right side of the paper,

|ext/north| will place a node towards the upper side of the paper and

|ext/south| will place a node towards the lower side of the paper.

The placement mechanisms |ext/left| and |ext/right| are like the original |left| and |right| mechanisms but don't swap si

Certain cases exist for |ext/west|, |ext/east|, |ext/north| and |ext/south| placements where it is not clear how a noc

Libraries These libraries (should) work with both and .

Utilities

Changelog, Index & References *Changelog 2Version (2026-07-26)

Bugfix to `|ext.positioning-plus|`. [?]

Bugfix to `|ext.beamer|`.

Bugfix to `|ext.paths.ortho|`.

Added node pictures to `|ext.nodes|`.

Added `|auto|` key to `|ext.topaths.arctthrough|`.

`|warn|` is not the default compatibility setting anymore.

Version 0.6.2 (2025-04-24)

Bugfix to and slight refactoring of `|ext.paths.ortho|`.

Bugfix to `|pgffor-ext|/|ext.misc|`.

Added better support for `|graphs|` library for `|ext.beamer|`.

Version 0.6.1 (2025-04-12)

Added library `ext.beamer`.

Added new tips `|ext_DoubleCap|`, `|ext_DoubleStealth|` and `|ext_DoubleTriangle|`.

Bugfix to `|ext.arrows-plus|`. [?]

Version 0.6 (2025-03-18)

Added `\tikzextset`, `\tikzextversion` and `\tikzextversionnumber`

Added six new `|auto|` placement mechanisms: `|ext/above|`, `|ext/below|`, `|ext/west|`, `|ext/east|`, `|ext/north|` and `|ext/south|`

Added `|ext/auto offset|` for `|auto|` placement.

Added `|ext/precise auto angle|`.

Added library `ext.arrows-plus`.

Added library `ext.topaths.autobend`.

Made `|ext.node-families|` and `|ext.scalepicture|` memoizable.

Version 0.5.1 (2023-04-02)

Added library `ext.arrows`.

Bugfix to `|ext.pgfkeys-plus|`. [?]

Version 0.5 (2023-03-17)

Added package `pgffor-ext`.

Added library `ext.nodes`.

Added library `ext.layers`.

Bugfixes to `|ext.calendar-plus|`.

Allow the original `|rectangle|` timer with `|ext.paths.timer|`.

Version 0.4.2 (2022-10-30)

Added library `ext.scalepicture`.

Bugfixes to `|shapes.uncenteredrectangle|`, `|paths.ortho|`, `|positioning-plus|` and `|pgfcalender-ext|`.

Version 0.4.1 (2022-10-23)

Cleaned up directory structure of documentary.

Added keys library `ext.pgfkeys-plus`.

Added shape `uncentered rectangle`
(library `ext.shapes.uncenteredrectangle`).

Fixed `|ext.paths.arcto|` – again [?].

Version 0.4 (2022-10-10)

CTAN version of 0.3.1

Version 0.3.1 (2022-10-09)

Fixed `|ext.paths.ortho|` keys `|only vertical first|` and `|only horizontal first|`.

Moved all (except the `|to path(s)|`) to namespace `|/tikz/ortho|`. `|/tikz/hvvh|` and `|/tikz/udlr|` are considered deprecated.

Fixed `|`.

Added more calendar tests.

Added directory structure.

Version 0.3 (2022-09-24)

Added shape `circle arrow`
(library `ext.shapes.circlearrow`).

Added shape `circle cross split`
(library `ext.shapes.circlecrosssplit`).

Added shape `heatmark`
(library `ext.shapes.heatmark`).

Added shape `rectangle with rounded corners`
(library `ext.shapes.rectangleroundedcorners`).

Added shape `superellipse`
(library `ext.shapes.superellipse`).

Added library `ext.node-families.shapes.geometric`.

Fixed `|ext.node-families|`’ key `|size|`.

Renamed internal macros to use custom namespace starting with `|@|`.

Added some references.

Version 0.2 (2022-08-21)

Added library `ext.positioning-plus`.

Added library `ext.node-families`.

Version 0.1 (2022-08-16)

Added library `ext.calendar-plus`.

Added library `ext.misc`.

Added library `ext.paths.arcto`.

Added library `ext.paths.ortho`.

Added library `ext.paths.timer`.

Added library `ext.patterns.images`.

Added library `ext.topaths.arctthrough`.

Added library `ext.transformations.mirror`.

Added library `ext.transformations.mirror`.